

# DATAAC102 - Data, Inference, and Decisions

## Overview

### Subject

DATA

### Course Number

C102

### Department

Data Science Undergraduate Studies

### Course Level

Undergraduate

### Formerly Known As

Statistics 102

### Course Title

Data, Inference, and Decisions

### Course Description

This course develops the probabilistic foundations of inference in data science, and builds a comprehensive view of the modeling and decision-making life cycle in data science including its human, social, and ethical implications. Topics include: frequentist and Bayesian decision-making, permutation testing, false discovery rate, probabilistic interpretations of models, Bayesian hierarchical models, basics of experimental design, confidence intervals, causal inference, Thompson sampling, optimal control, Q-learning, differential privacy, clustering algorithms, recommendation systems and an introduction to machine learning tools including decision trees, neural networks and ensemble methods.

### Units

#### Minimum:

4

#### Maximum:

4

### Grading Basis

Default Letter Grade; P/NP Option

### Reading and Composition Requirement

None of the Reading and Composition Requirement

### Simple Requirements

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| <div> <div>Course Enrollment Requirements</div> <div> <div>Type</div> <div>Prerequisite</div> <div> <div>Requirement</div> <div> Math 54 or 56 or 110 or Stat 89A or Physics 89 or both of ELENG 64 and ELENG 66 with a C- or better, or Pass; Data/Stat/CompSci C100 with a C- or better, or Pass; and any of EECS 126, Data/Stat C140, Stat 134, IndEng 172, Math 106 with a C- or better, or Pass. Data/Stat C140 or EECS 126 are preferred </div> </div> </div> </div> |
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### Repeat Rules

Course is not repeatable for credit.

### Credit Restriction Courses. Students will receive no credit for this course if the following the course(s) have already been completed.

STAT102 - Introduction to the Theory of Statistics (Inactive), DATAAC102 - Data, Inference, and Decisions, STATC102 - Data, Inference, and Decisions

### Cross-Listed Course(s)

STATC102 DATA INF DEC

### Formats

Discussion, Laboratory, Lecture

### Durations and Hours

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| <div> <div>Term</div> <div>Fall and Spring</div> <div> <div>Outside Work</div> <div> </div> </div> </div> |
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